

SKILL 3 - FEATURE ENGINEERING AND EDA

One Piece Bounty Dataset

Rows: 165 | Known bounties: 145 | Unknown: 20

Columns: 3 raw -> 23 engineered

Tier rules

Calamity : bounty $\geq 1,000,000,000$

Menace : 300,000,000 to 999,999,999

Wanted : below 300,000,000

Feature groups

Scaling : Billions, Log10, Percentile, Z-score, Min-Max

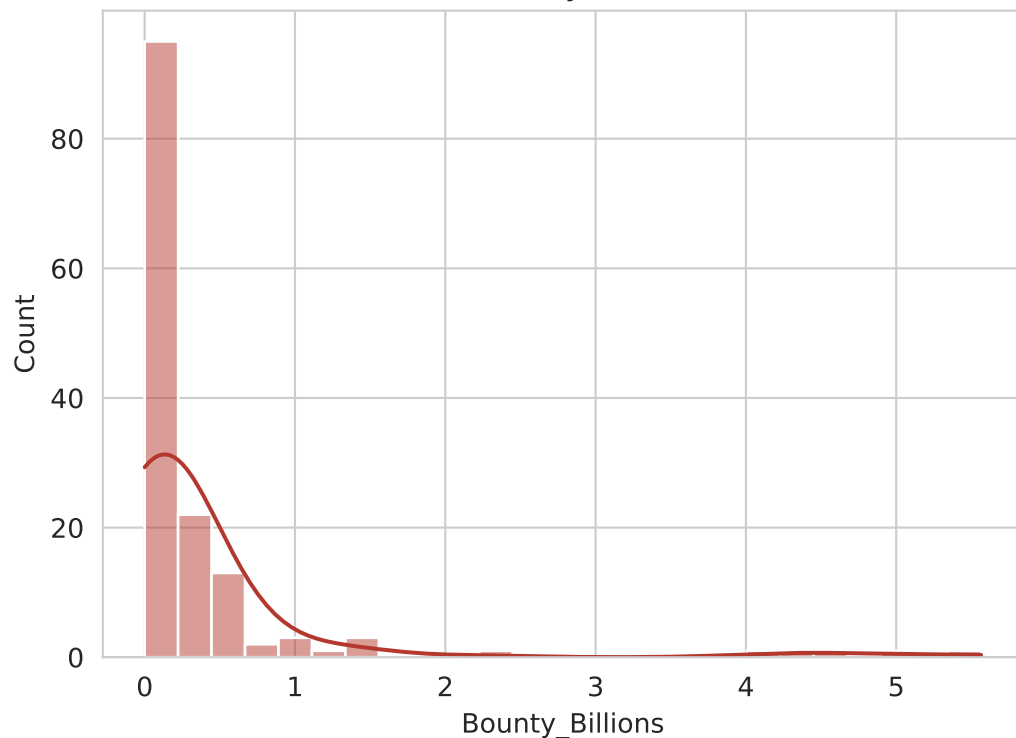
Binning : Tier, Tier_Code, one-hot Tier columns

Text : Name_Length, Word_Count, Has_D_Clan, Has_Parentheses, Is_Anonymous

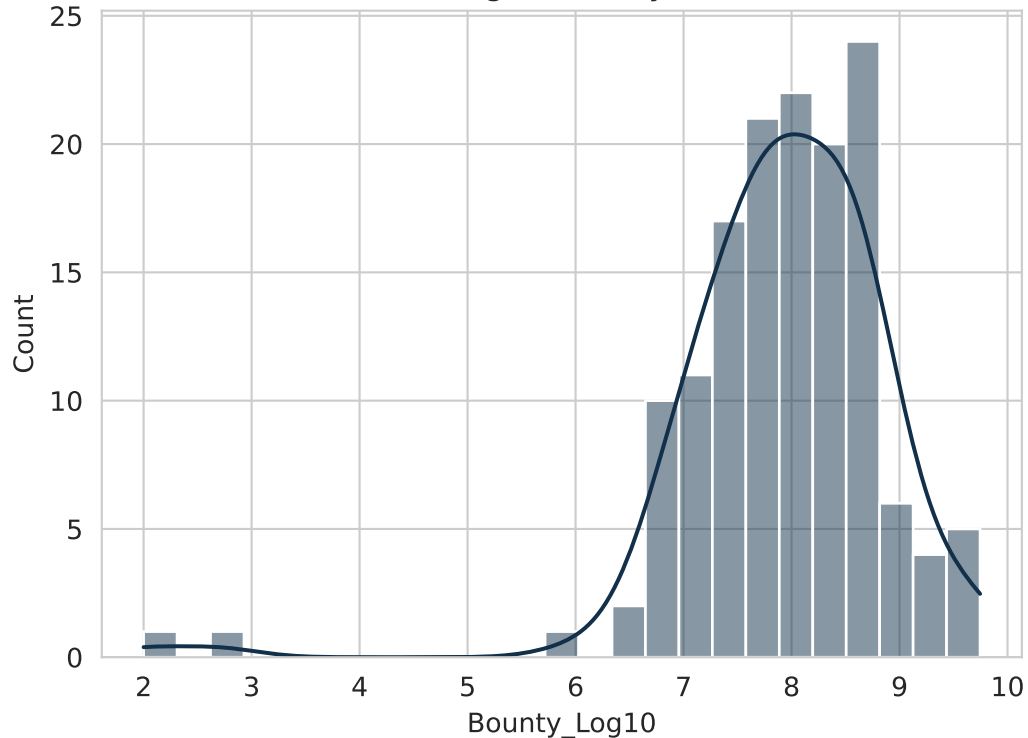
Grouping : Name_Group, Group_Size (frequency encoding), Is_Charlotte

Target : High_Bounty (log bounty $\geq$ median)

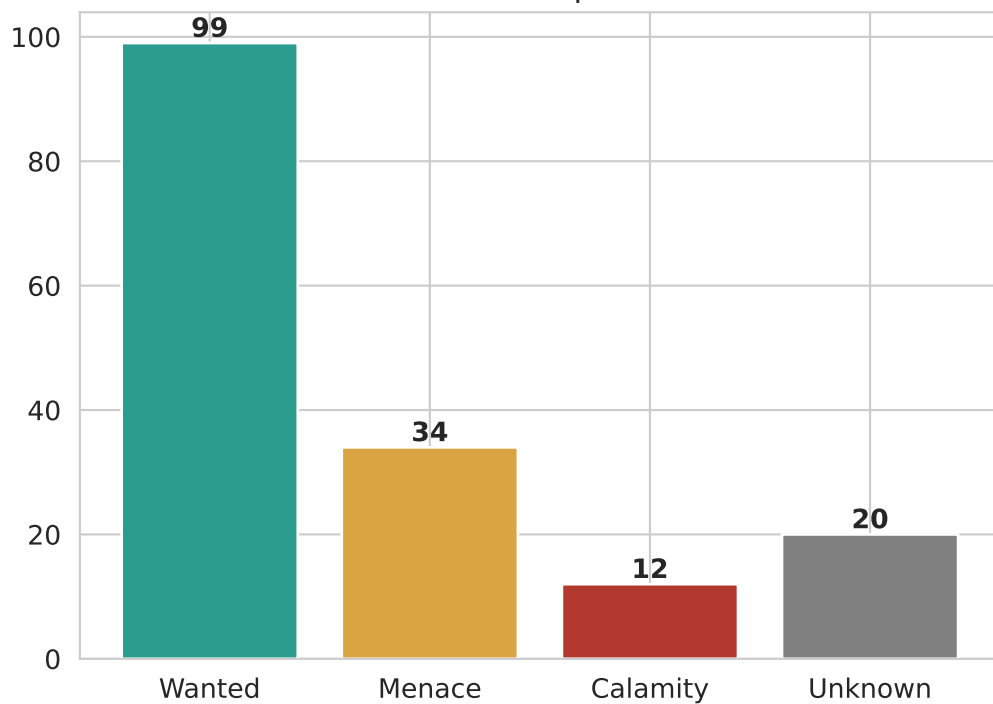
Raw bounty (billions)



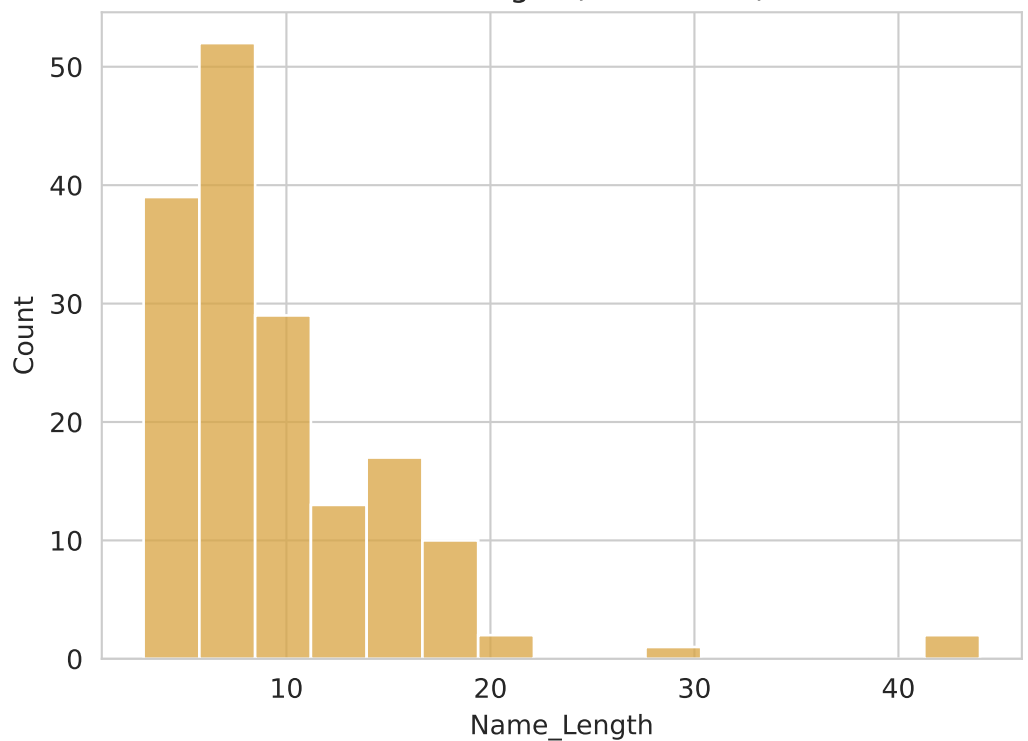
log10 bounty



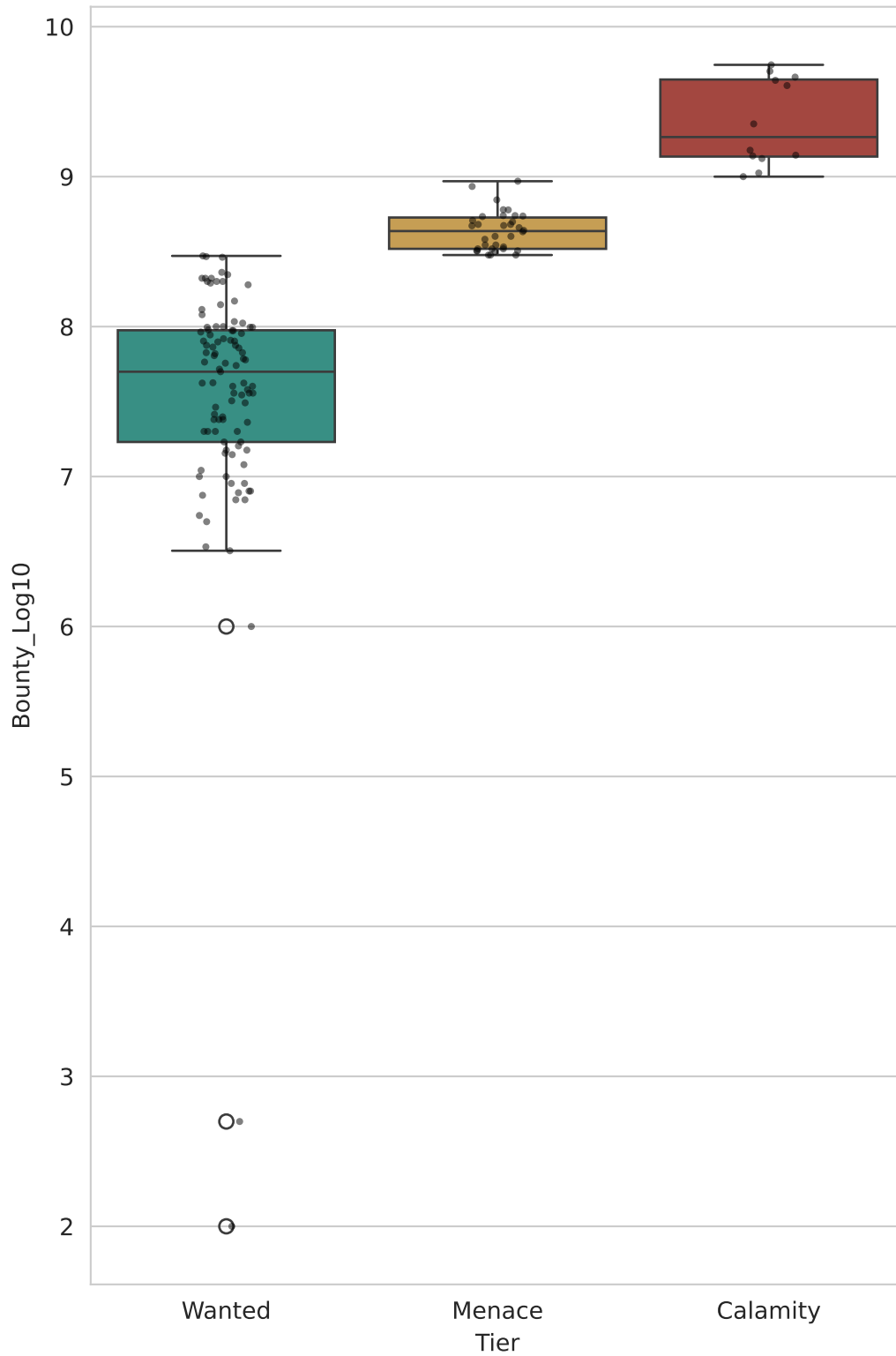
Characters per tier



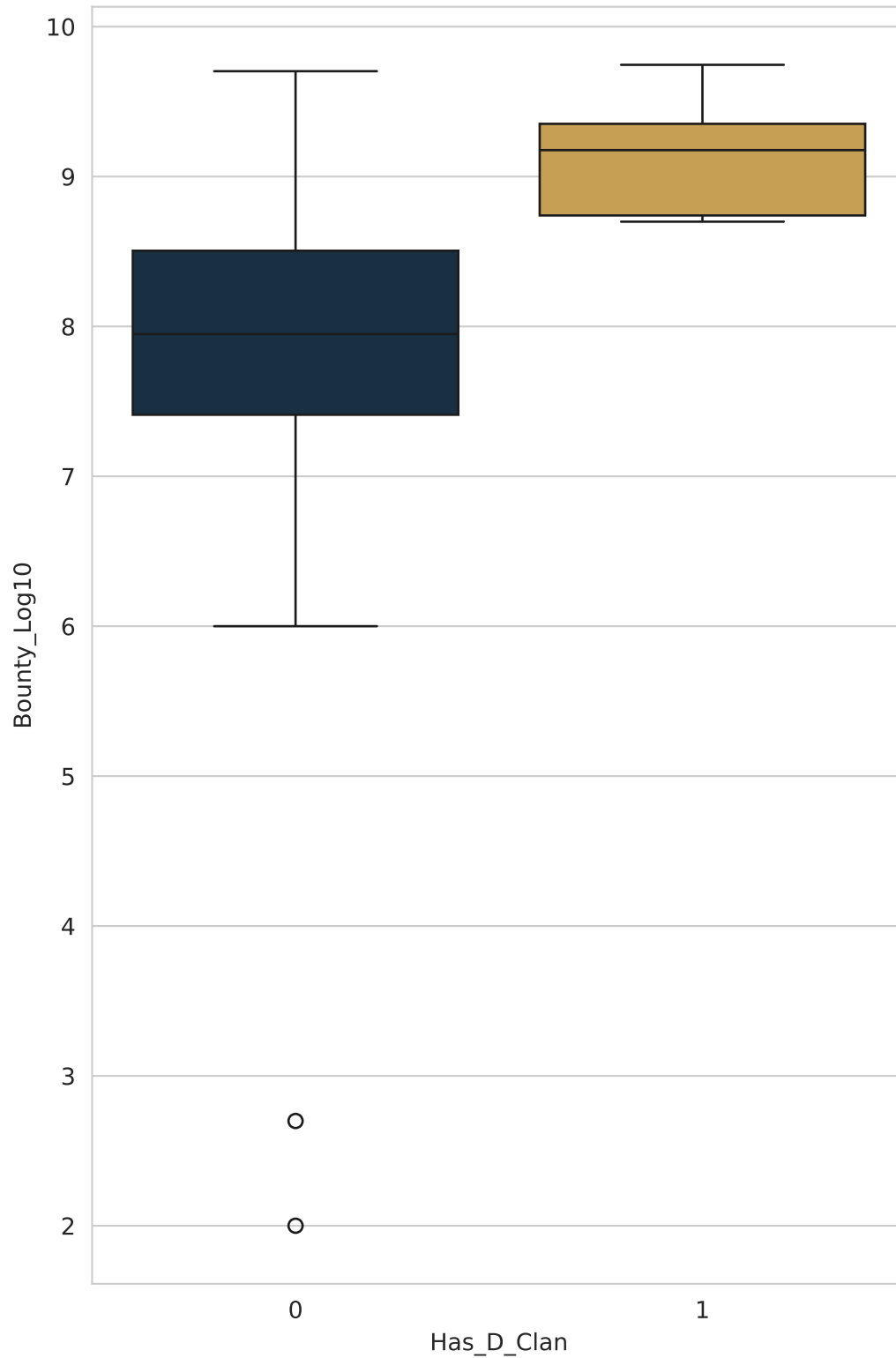
Name length (characters)



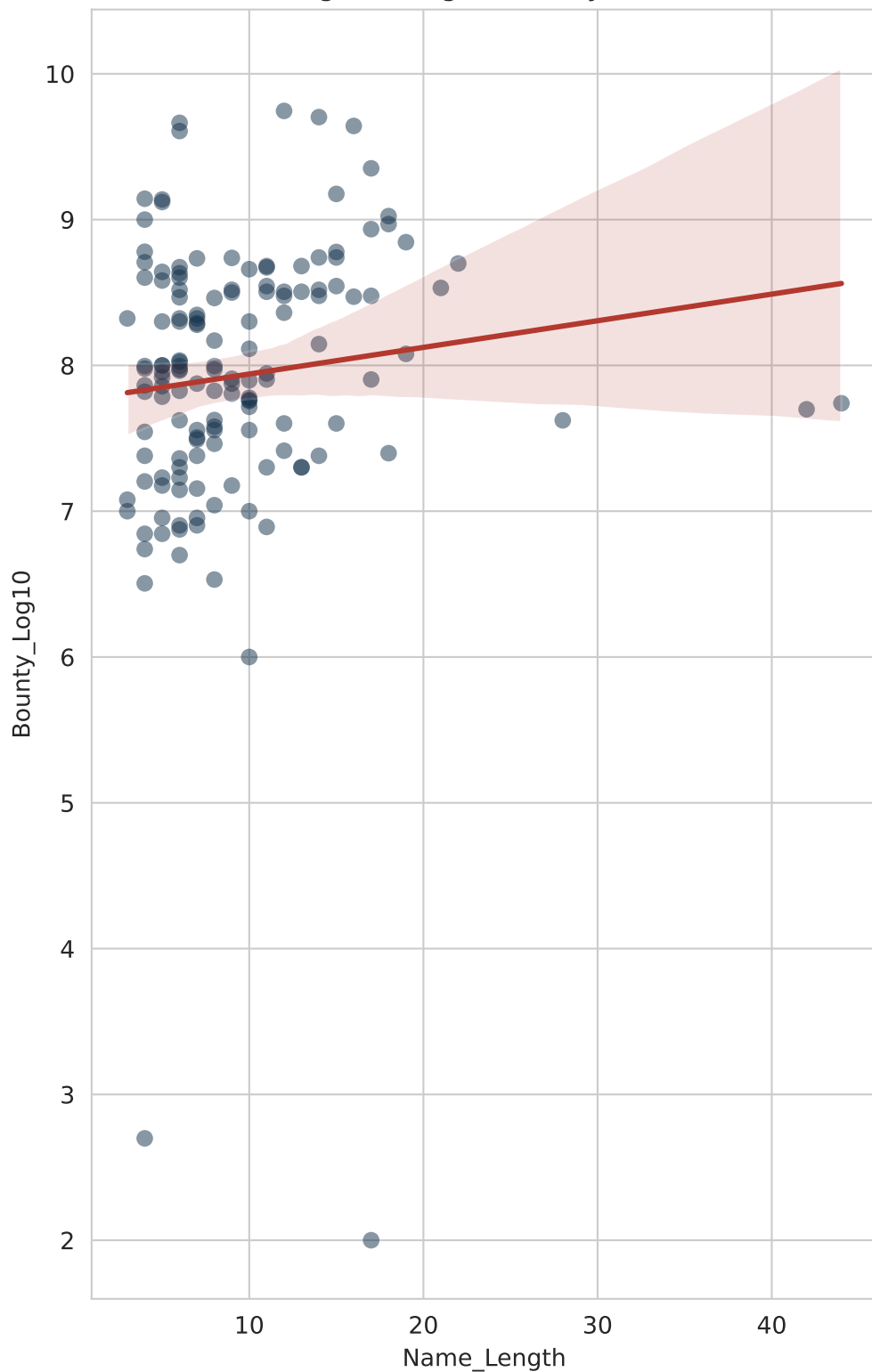
log10 bounty by tier



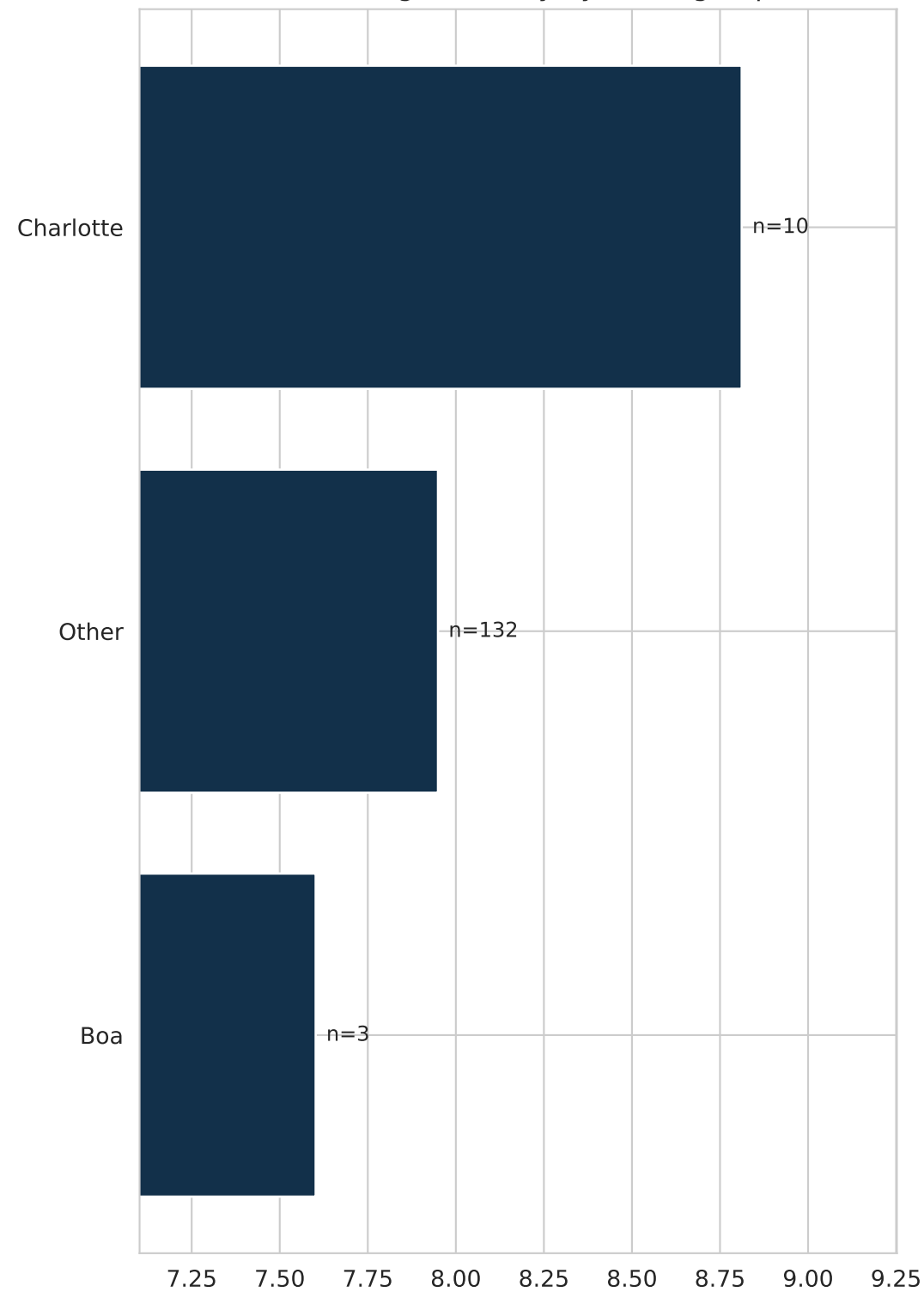
log10 bounty: D. clan (1) vs rest (0)



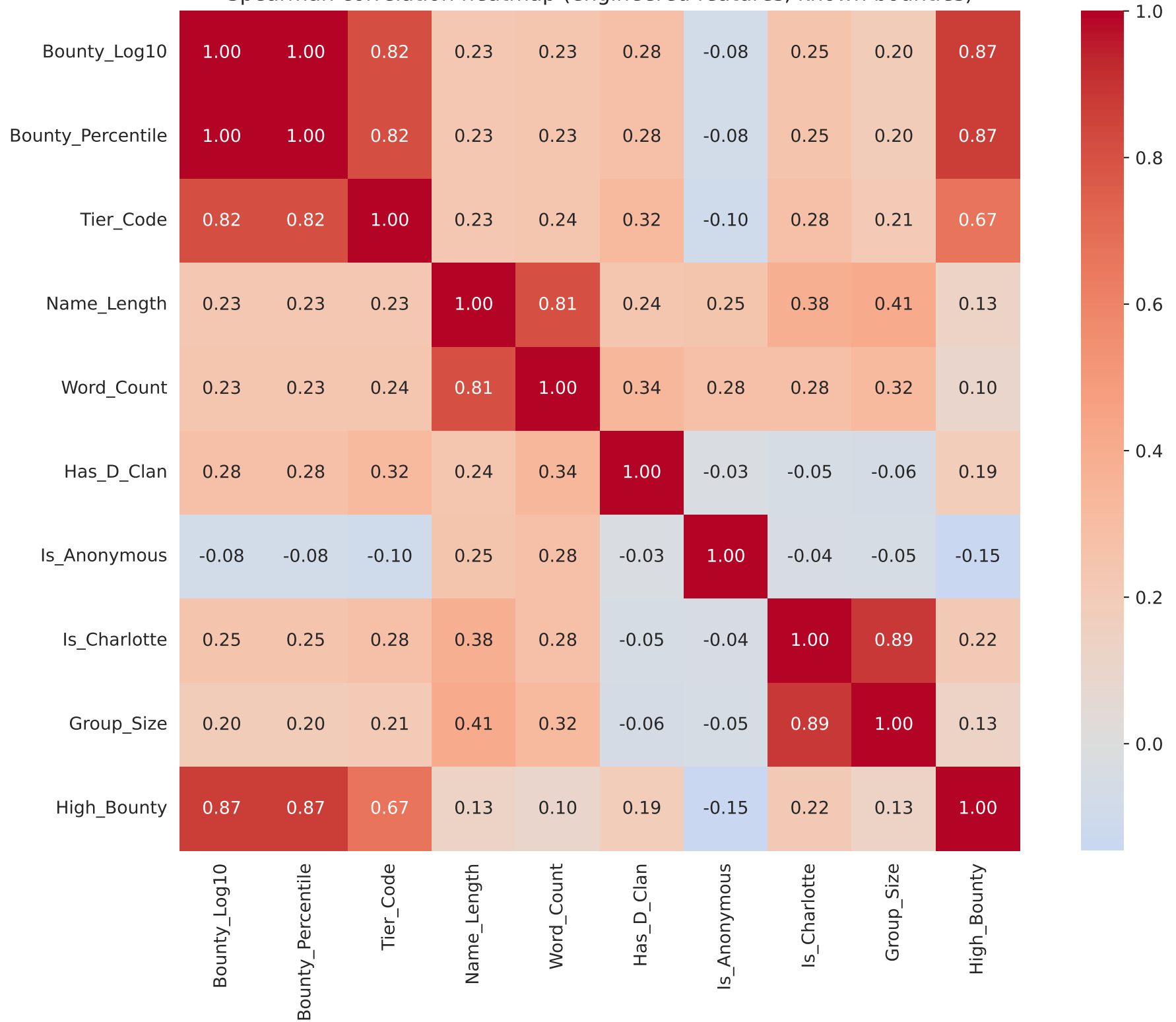
Name length vs log10 bounty (rho = 0.23)



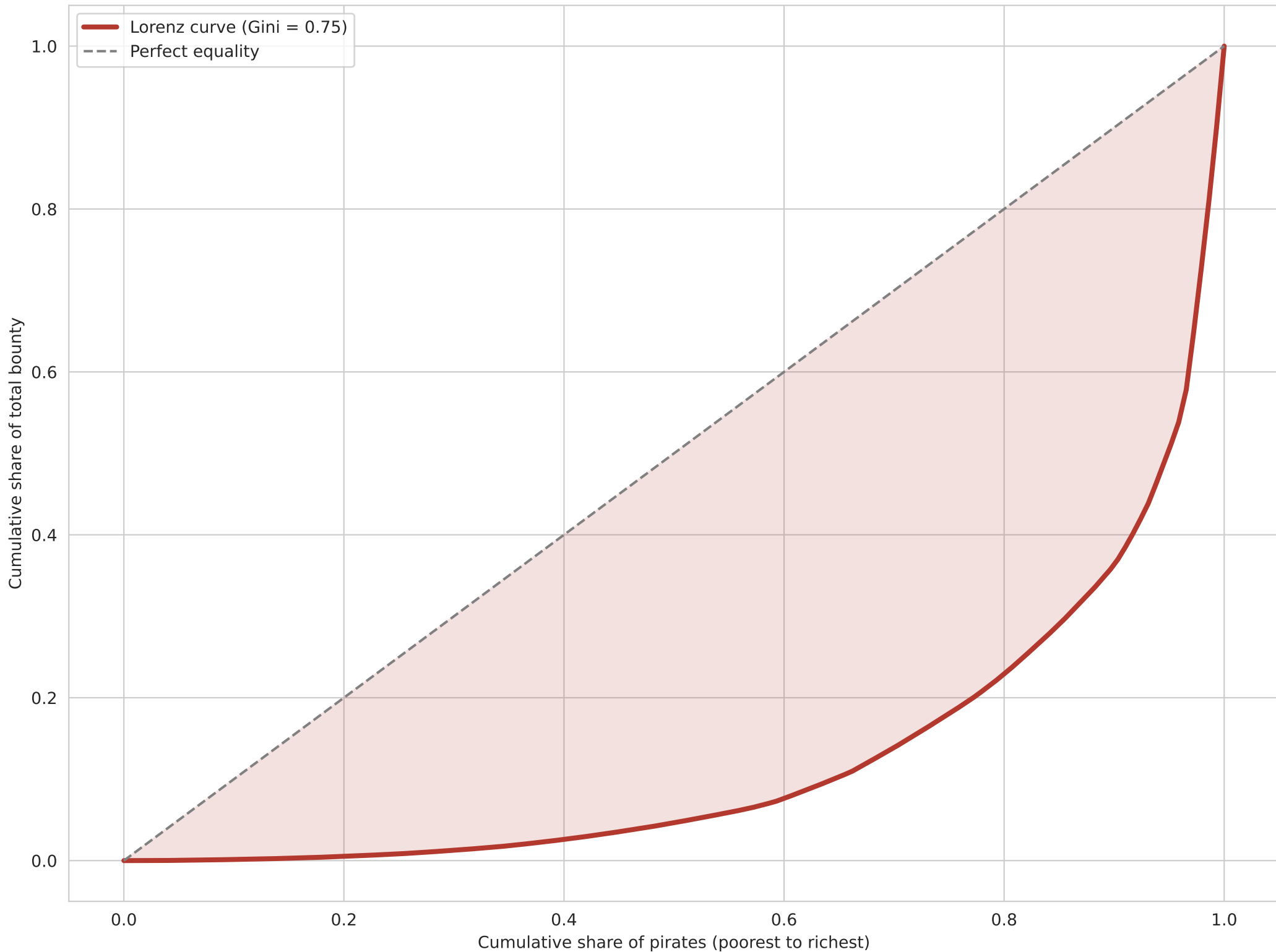
Median log10 bounty by name group



Spearman correlation heatmap (engineered features, known bounties)



Bounty inequality: top 5 pirates hold 42.2% of all known bounty



FEATURE DICTIONARY

Bounty_Billions	float64	nulls=20
Bounty_Log10	float64	nulls=20
Bounty_Percentile	float64	nulls=20
Bounty_Log_Zscore	float64	nulls=20
Bounty_Log_MinMax	float64	nulls=20
Tier	object	nulls=0
Tier_Code	float64	nulls=20
Tier_Calamity	int64	nulls=0
Tier_Menace	int64	nulls=0
Tier_Unknown	int64	nulls=0
Tier_Wanted	int64	nulls=0
Is_Anonymous	int64	nulls=0
Name_Length	int64	nulls=0
Word_Count	int64	nulls=0
Has_D_Clan	int64	nulls=0
Has_Parentheses	int64	nulls=0
Name_Group	object	nulls=0
Group_Size	int64	nulls=0
Is_Charlotte	int64	nulls=0
High_Bounty	float64	nulls=20